Determinants of Gold Price Volatility in Response to the Release of Non-Farm Payrolls: An Econometric Analysis

Abstract

This study analyzes the determinants of gold price volatility in the five minutes preceding and following the release of Non-Farm Payrolls (NFP), using an econometric approach. The NFP is a key indicator of the U.S. labor market, released monthly by the Bureau of Labor Statistics, and is often considered a catalyst for movements in financial markets. Volatility is modeled based on three main variables: weekly average volatility, the gap between expected and actual NFP values, and the presence of significant geopolitical events, identified through a dummy variable based on interquartile analysis.

The initial model highlighted the presence of autocorrelation in the residuals, which was addressed through a model transformation. The results reveal interesting dynamics in gold price movements in response to NFP releases, suggesting valuable insights for further exploration and applications in quantitative strategy and investment. This study represents a first step toward a more detailed understanding of the interactions between financial markets and globally significant macroeconomic events.

The Econometric Model

$$\text{price\_range} = \beta_0 + \beta_1 \cdot \text{volatility\_mean\_week} + \beta_2 \cdot \text{dummy\_geopolitics} + \beta_3 \cdot \text{forecast\_error} + \epsilon$$

Where:

- **price_range**: price range of XAUUSD in the 5 minutes surrounding the NFP release (dependent variable).

- **volatility_mean_week**: average weekly volatility.
- **dummy_geopolitics**: binary variable indicating the presence of significant geopolitical events.
- **forecast_error**: difference between the expected and actual values of the NFP.
- ϵ: error term.